2. After-Tax Returns

I start by showing the effect of taxes on savings vehicles with different tax treatments. I adapt the treatment in chapter 3 of Scholes et al. (2005), a marvelous book on *Taxes and Business Strategy* for the MBA classroom. Myron Scholes, the Nobel Prize winner and primary author of the book, asserted not very convincingly, "I said I was not an expert with regard to taxes," under cross-examination by the Justice Department in 2003 over an abusive tax shelter associated with his hedge fund, Long-Term Capital Management (LTCM). Scholes and other LTCM partners paid $40 million in fines and penalties associated with the transaction.[2] LTCM blew up in 1998 and was bailed out by a consortium of banks coordinated by the Federal Reserve Bank of New York. Scholes's second hedge fund after LTCM, Platinum Grove Asset Management, was forced to halt investor withdrawals (it invoked *gates*, see chapter 17) after losing close to 40% of its value in 2008 during the financial crisis.[3]

In our examples below, we assume an income tax rate of 39.6%, a capital gains rate of 23.8%, and that each asset earns a constant annual return of 12%. We start with $1,000 invested in each asset in year zero. Consider the following investments:

Bond Fund: where all distributions are taxed as ordinary income and there are no capital gains or losses. After n years, $1,000 initially invested in the bond fund cumulates to

$$1,000 \times (1 + 0.12(1 - 0.396))^n, \tag{12.1}$$

where every year the investor can only keep 60.4c of each dollar earned by the fund. This is the lowest after-tax accumulation of all the strategies we consider.

Non-Deductible IRA: where contributions are not tax deductible (that is, contributions are made with after-tax dollars) but earnings are not taxed until retirement, at which point they are taxed as income. In this case, the non-deductible IRA cumulates to

$$\underbrace{1,000 \times (1 + 0.12)^n}_{\text{Pre-tax accumulation}} - \underbrace{0.396[1,000 \times (1 + 0.12)^n - 1,000]}_{\text{Taxed at withdrawal as income}}. \tag{12.2}$$

Compared to the bond fund, the non-deductible IRA accumulates at the pre-tax return of 12% rather than the after-tax return of $(1 - 0.396) \times 0.12 = 7.2\%$.

[2] Johnston, D. C., "Economist Questioned on Tax Shelter Role," *New York Times*," July 10, 2003. Browning, L., and D. C. Johnston, "Hedge Fund Is Censured and Ordered to Pay Taxes," *New York Times*, Aug. 28, 2004.

[3] Kishan, S., "Scholes's Platinum Grove Fund Halts Withdrawals after Losses," Bloomberg, Nov. 6, 2008.